# MEIOSIS

- It is the type of cell division in which one matured and diploid mother cell produces four daughter cells; each having half number of chromosome than mother cell.

So, it is called as **Reductional cell division**.

It was first studied by Van  Benden and term given by Farmer and Moore.

Meiosis takes place in reproductive cells or germ cells. E.g. Testis ,ovaries in animals , pollen mother cells of anther and megaspore mother cels of in the ovule of flower in plants.

The cells which divide by meiosis are called as Meiocytes.

1. Meiosis completes in two phases

A. Meiosis-I

B. Meiosis- II

a. Meiosis –I

The meiosis –I is also known as reductional division or heterotypic division as it involves the formation of two daughter cells with half number of chromosomes as that present in the mother cell.

It is completed in

1. Interphase
2. Karyokinesis –I
3. Cytokinesis -I

# Interphase

It is similar to that og mitosis ,i.e. having G1, s and G2 phases.

Karyokinesis I

It is division of nucleus and completes in four phases

i. Prophase I
ii. Metaphase I
iii. Anaphase I
iv. Telophase I

# Prophase I

It is the longest and a complex phase.

It completes in following five sub phases

i. Leptotene
ii. Zygotene
iii. Pachytene
iv. Diplotene
v. Diakinesis

# Leptotene

It is also called as thin thread stage. It involves the following events

i. The size and volume of nucleus increases.
ii. Condensation, shortening and thickening of chromosome takes place.
iii. Chromosome appeared as thread like single stranded.
iv. Chromosomes have beaded structures along the entire length and are called as Chromomeres.
v. Nuclear membrane and nucleolus remain intact.

# Zygotene

i. The chromosomes become more distinct by their further shortening, thickening and coiling.
ii. Pairing of homologous chromosomes (one from paternal and one from maternal) occurs to form bivalents. The process of pairing is called as synapsis.
iii. A nucleo protein complex called as synaptonemal complex develop between the synapsed homologous chromosomes.

# Pachytene

i. The bivalents become shorter and thicker and more distinct.

ii. The chromatids of each bivalent slightly separate and become visible. The two visible chromatids of a chromosome are called as Dyad. A group of four homologous chromatids (two dyads) is called as Tetrad.

iii. The two chromatids of the same chromosome are called sister chromatids and those of dissimilar chromosomes are called non sister chromatids.

iv. Crossing over(recombination)occurs during Pachytene.

Recombination involves mutual exchange of the corresponding segments of non sister chromatids of homologous chromosomes

v. The sites where crossing overtakes place are called as Chiasmata.

vi. The nucleolus remains prominent.

# Diplotene

i. The homologous chromosomes start separating.This is called disjunction.
ii. The separation of chromosomes does not take place at the points called as chiasmata.
iii. The nuclear membrane remains intact, but the nucleolus tends to degenerate.

**Diakinesis**

i. Chiasmata are shifted towards the terminal end of chromosomes which is called as terminalization of chiasmata.
ii. Nuclear membrane begins to disappear.
iii. nucleolus disappear completely . Spindle fibers begin to appear.

# Metaphase I

i. Nuclear membrane disappear completely and develops the spindle apparatus.
ii. The spindle fibers converse towards the two ends called poles.
iii. The bivalents are arranged at the equator in double metaphasic plate.
iv. The arms of chromosomes lie towards the equator, but centromeres lie towards the poles.
v. centromere attaches with spindle fiber at tractile fiber.

# Anaphase I

i. Homologous chromosomes break their connections and separate out which is called disjunction.
ii. Chiasmata disintegrate. There is no division of centromeres. So, chromosomes are double stranded.
iii. The chromosomes move towards two opposite poles.
iv. At the end of anaphase I, two groups of chromosomes are produced, with each group having half the number of chromosomes.

# Telophase I

i. Nuclear membrane  and nucleolus reappear, and spindle fibers disappear.
ii. Chromosomes undergo decongestion . They elongate and form chromatin reticulumn.
iii. At the end of telophase I two haploid nuclei are formed.

Telophase I  may or may not followed by cytokinesis.

If cytokinesis takes place, two haploid cells are formed.

# Cytokinesis I

In meiosis cytokinesis is of two types : successive and simultaneous.

In simultaneous type, cytokinesis I is omitted while it occur in successive types where cytokinesis occurs after both meiosis I and Meiosis II.

Cytokinesis commonly take place by cleavage in both plants and animals.

Cytokinesis I produces two daughter cells with nuclei having haploid number of chromosomes.

# *Interkinesis*

It is a short interphase rarely present in between meiosis I and Meiosis II.

Biomolecules synthesize here but DNA replication does not take place.

Meiosis II

It is also called as equational or homotypic cell division because of maintaining same number of chromosomes as after meiosis I.

It completes in two steps: Karyokinesis II and Cytokinesis II.

Karyokinesis II

It completes in four phases: Prophase II , Metaphase II ,Anaphase II and Telophse II.

# Prophase II

1. The chromatin fibres shorten and thicken.
2. The nucleolus degenerates.
3. Nuclear membrane start to degenerate and spindle fibre begin to appear.
4. In animal cell the centriole divides into two and poles.

Metaphase II

1. Nuclear membrane  and nucleolus completely disappear.
2. Spindle fibres are are completely developed.
3. Tactile fibres connected at kinetochore of each chromosome.
4. All chromosomes lie on the equatorial plate.
5. The centromere of all chromosomes lie on the equator  but arms of chromosomes are placed variously.

# Anaphase II

1. The centromere of each chromosome divides and forms two daughter centromeres.
2. Each double stranded  chromosome changes into two daughter single stranded chromosomes.
3. These daughter chromosomes move towards the opposite poles due to contraction of tractile fibres.
4. Centromeres of chromosomes are faces towards the poles but arms of chromosomes are faces towards the equatopr.
5. At the end of anaphase II, each poles consists of haploid number and single stranded chromosomes.

# Telophase II

1. Chromosomes lie in two opposite poles. They undergo condensation.
2. They elongate to form chromatin reticulum.
3. Nucleolus and nuclear membrane reappears.
4. There is formation of four haploid daughter nuclei.
5. Spindle fibers completely disappear.
6. Two centrioles' pair themselves into centrosomes.

# Cytokinesis II

In successive type of cytokinesis, both the daughter cells undergo independent cytokinesis after the completion of karyokinesis I as well as karyokinesis II by cleavage (as in linear or isobilateral tetrad)

In simultaneous type, cytokinesis is absent after meiosis I.

All the haploid nuclei formed after meiosis II undergo cytokinesis by cleavage method and form four haploid cells (as isobilateral tetrad).

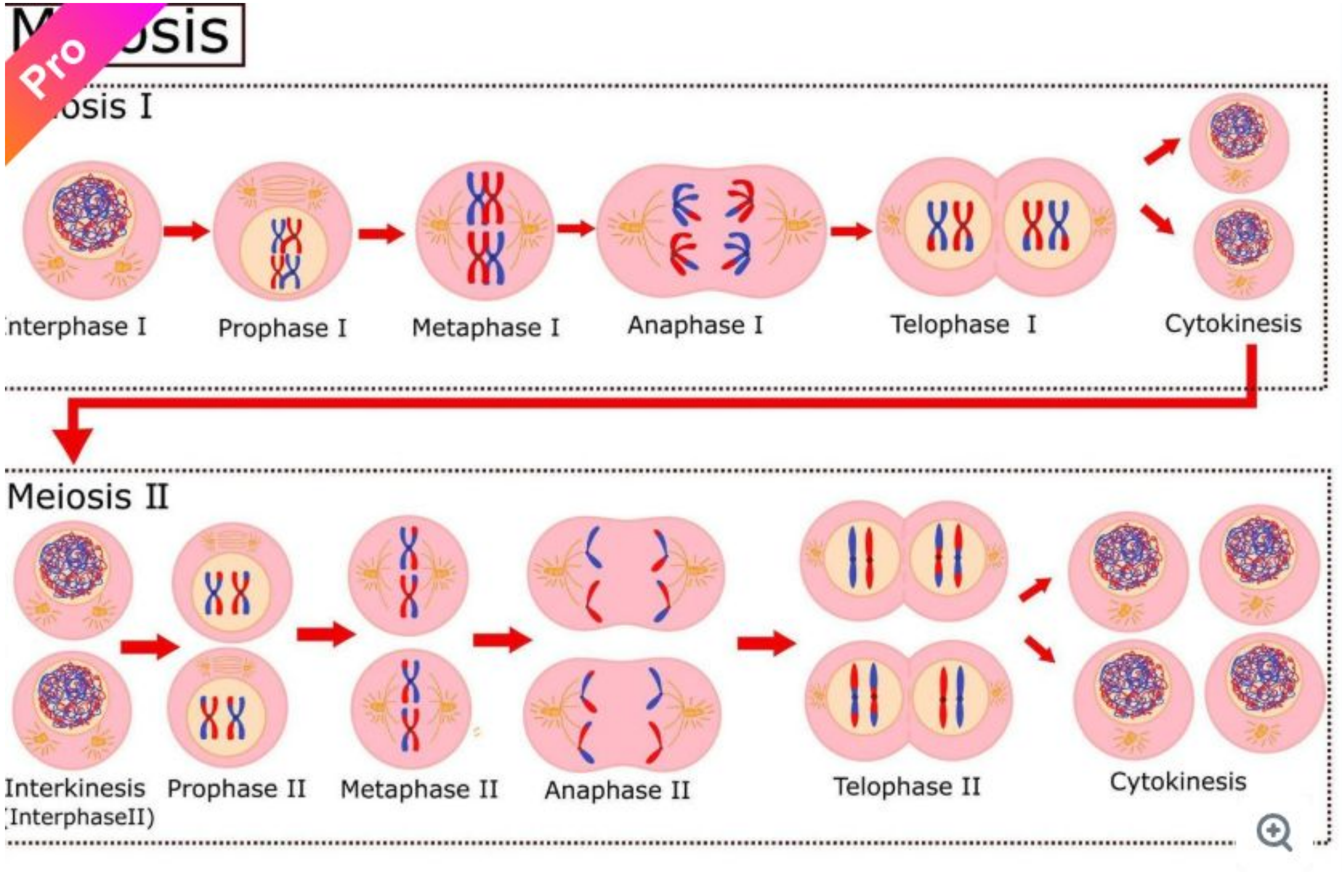
# Significances of meiosis

1. It is essential for life cycle of sexually reproducing organisms because it helps in the formation of haploid gametes.
2. Meiosis produces the haploid gametes by reducing the diploid number of parental chromosomes . These gametes later fuse by the process of fertilization and form diploid structures. So, it maintains the fixed number of chromosomes in generation to generations.
3. The law of segregation and law of independent assortments are based on meiosis and fertilization.
4. Crossing over occurs in meiosis that helps in the formation of new characters in organisms which is raw material for evolution as well as for better adaptability and resistivity of organisms.
5. Chromosomal and genematic mutations can takes place by irregularity of meiotic divisions.
6. Meiosis is useful to know the phylogenetic and evolutionary trend in different organisms.

7. The haploid gametes are produced due to meiosis and after fusion of opposite gametes a diploid zygote is formed which develop in sporophyte. So, alternation of generation is based on meiosis.

8. Mutation may occur due to irregularity in meiosis.

| Mitosis | Meiosis |
|---|---|
| 1) It occurs in somatic cells. | 1) It occurs in germ cells. |
| 2) Nucleus divides only once. | 2) Nucleus divides twice. |
| 3) Two daughter cells are formed. | 3) Four daughter cells are formed. |
| 4) Daughter cells are diploid. | 4) Daughter cells are haploid. |
| 5) It occurs more frequently. | 5) It occurs less frequently. |
| 6) Daughter cells form somatic organs. | 6) Daughter cells form gametes. |
| 7) There is only one prophase, one metaphase one anaphase and one telophase. | 7) There are two of each phase and five sub-phases in prophase - 1. |
| 8) Number of chromosomes are not changed in the daughter cells. | 8) Number of chromosomes are reduced to half. |
| 9) Chromosome number doubles at the beginning of each cell division. | 9) Chromosome number is not doubled. It doubles after the end of first meiotic division. |
| 10) No crossing over in chromosomes. | 10) Crossing over occurs chromosomes. |
| 11) Equation division. | 11) Reduction division. |


Meiosis
Pro
Meiosis I
Interphase I
Prophase I
Metaphase I
Anaphase I
Telophase I
Cytokinesis
Meiosis II
Interkinesis
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Prophase II
Metaphase II
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Cytokinesis

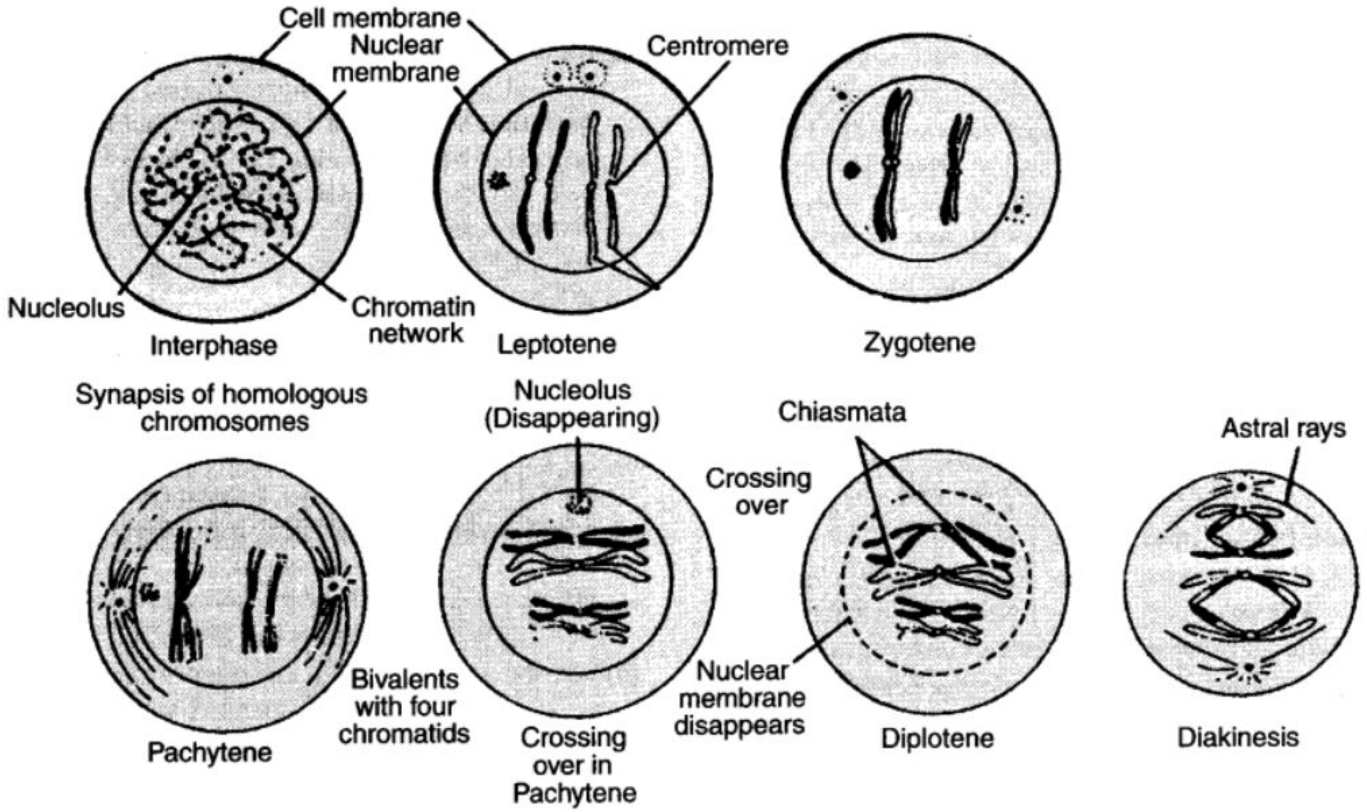


**Fig. : Prophase I of meiosis I**